Journal of Electronic Testing (2026) 42:57–75
https://doi.org/10.1007/s10836-026-06215-8

RESEARCH

Impedance Side-Channel Analysis of ASICs: An investigation of measurement factors

Bijan Fadaeinia[1] · **Shahin Tajik**[2] · **Amir Moradi**[3]

Received: 4 March 2025 / Accepted: 31 December 2025 / Published online: 21 February 2026
© The Author(s) 2026

Abstract

A substantial body of research has been conducted on the subject of physical side-channel analysis attacks and the measures that can be employed to counteract them. These attacks typically exploit the impact of computation or storage on current consumption or voltage drop on a chip, which is an unavoidable consequence of the underlying physical processes. This data-dependent influence can be exploited through analytical techniques such as power or electromagnetic analysis. Recently, a novel target for side-channel analysis adversaries has emerged, based on the data dependency between the chip's power delivery network impedance and the temporarily stored content in registers. There are two principal reasons why information leakage through the Impedance Side Channel (IMSC) compromises the security of the implementations. First, this method can target the secret even outside the time window, provided that the sensitive data is stored somewhere in the circuit; and second, the simultaneous and independent probing of particular registers challenges the t-probing security model used in masking proofs, a primary side-channel countermeasure. It is crucial to highlight that the interdependence between the die impedance and the temporarily stored data in registers is revealed through scattering parameter analysis. Consequently, the precise measurement of data-dependent impedance, or equivalently, the Scattering (S)- parameter, is essential for its use as a side-channel, which is our focus in this study. In this study, we examine the impact of environmental factors on the IMSC, which are controlled by a classical side-channel adversary. Such factors include temperature and supply voltage. Due to the similarity between the measurement procedures of IMSC and Static Power Side- Channel Analysis (SPSCA), we further provide a fair comparison of these two side-channels in terms of their exploitability and ease of measurement. Most of the previously published studies on the IMSC have been conducted using Field Programmable Gate Arrays (FPGAs). This provides a high level of control over the placement of design components, which can, in turn, affect analysis results. In this work, we use a dedicated Application-Specific Integrated Circuit (ASIC) chip fabricated in 28nm Complementary Metal-Oxide-Semiconductor (CMOS) technology to conduct our study in a more realistic setting. We demonstrated the significant impact of the aforementioned environmental factors on the exploitability of such a side channel. In conclusion, an IMSC adversary can influence the device to leak more information by regulating its operational environment.

Keywords Impedance side-channel analysis · Static power · ASIC · Temperature · Supply voltage · Scattering parameter · Vector network analyzer · Source meter

Shahin Tajik and Amir Moradi contributed equally to this work.

Responsible Editor: K. J. Lee

✉ Bijan Fadaeinia
bijan.fadaeinia@rub.de

Shahin Tajik
stajik@wpi.edu

Amir Moradi
amir.moradi@tu-darmstadt.de

[1] Ruhr University Bochum, Bochum, Germany

[2] Worcester Polytechnic Institute, Worcester, MA, USA

[3] Darmstadt University of Technology, Darmstadt, Germany

1 Introduction

It is crucial to implement robust encryption to safeguard the security and integrity of data handled in embedded systems. While cryptographic primitives provide mathematical security assurances to increase confidence in their resistance against modern computational attacks, these protections can still be compromised by physical attacks [1–3]. Attackers who have physical access to devices and are able to measure the physical emissions of cryptographic circuits during the execution of a primitive can carry out physical attacks. These attacks, with their potential to undermine the entire security framework despite the advanced cryptographic countermeasures in place, highlight the urgency of the issue.

Physical attacks, including Side-Channel Analysis (SCA) and Fault-Injection (FI), often involve adversaries tampering with systems at various levels, from Printed Circuit Boards (PCBs) to Integrated Circuits (ICs) [4]. For instance, in power analysis attacks, the adversary may modify the PCB by soldering or replacing a shunt resistor on the power rails or removing decoupling capacitors to amplify leakage detection through power consumption [5]. These modifications enable the attackers to correlate such SCA observations with the intermediate values of the underlying cryptographic algorithm, thereby compromising the security of virtually all unprotected cryptographic primitives. As a result, even mathematically secure primitives can be undermined by analyzing the data-dependent fluctuations of the device through physical emissions during execution. These data-dependent fluctuations manifest through various measurable quantities, such as dynamic power consumption [1], static power consumption [6], electromagnetic emanation [1], acoustic waves [7], photon emission [8], and thermal radiation [9]. Over the past decade, these quantities have been exploited in various SCA attacks to break the security of numerous cryptographic implementations. Concurrently, various counter measures, such as hiding [10, 11] and masking [12], have been developed to mitigate these attacks. It is crucial to implement dedicated measures that minimize information leakage through physical side channels to maintain the security of cryptographic systems. The role of the aforementioned side channels has been extensively studied over the years. Recently, a new side channel has been introduced that explores the dependency between the chip's (die's) impedance and the intermediate values of the underlying cryptographic algorithm [13]. This novel approach, emerging in the past year, reveals another dimension of vulnerability in cryptographic implementations, further emphasizing the need for robust and comprehensive security measures.

1.1 Related Works

Recently, a novel backscattering side-channel method for chip-level tamper and counterfeit detection has been presented, additionally introducing a fresh perspective to the existing non-invasive verification techniques. Unlike previous EM backscattering approaches, which inject EM waves at a specific frequency into the chip and primarily focus on switching activity [14–17] of transistors, this method identifies alterations in a chip's impedance spectrum due to tampering [18, 19]. This method encompasses modifications, including changes in the logic elements, placement, routing, and defects. The underlying principle measures the echo of injected high-frequency signals to determine impedance changes induced by physical alterations. A Radio Frequency (RF) network analyzer injects high-frequency sine waves and measures the resulting reflections, quantifying impedance changes using the so-called Scattering Parameters (*S*-parameters). The resulting reflection response provides insights into the tamper event based on its impact on the chip's die. Even minor physical alterations significantly impact a chip's impedance, resulting in detectable changes in the reflection response. By analyzing the impedance spectrum, the method accurately identifies the nature and extent of the tamper event. Along the same line and based on the findings of [18], the technique has been extended to detect physical tampering in cryptographic devices using a self-contained tamper-evident sensor [4]. This method uses impedance characterization of the Power Distribution Network (PDN) for tamper detection. For impedance characterization, a circuit-based sensor composed of a Ring-Oscillator (RO)-based Analog-to-Digital Converter (ADC) is deployed to measure the impedance value of the PDN by measuring the frequency changes in an RO. Afterward, the information leakage through the impedance of an integrated circuit was presented as a potential security threat for cryptographic hardware [13, 20, 21].

The authors of [13] investigated the data dependency of the impedance of a CMOS-based circuit in Digilent Arty S7 development boards [22], which hosts an AMD/Xilinx Spartan-7 FPGA manufactured using 28nm technology. The authors demonstrated that the data stored in an FPGA register changes the impedance profile, which can be measured across different frequency bands. Additionally, registers located in different places affect the impedance profile at different frequency bands. They also found out that longer routes in FPGA designs could increase leakage, making them more vulnerable to SCA attacks through the impedance[1]. The feasibility of such attacks was demonstrated in practice, and as an SCA attack, the differential analysis was

[1] We should highlight that long routes in FPGAs would go through multiple switch boxes, which are also realized by combinational circuits.

performed on frequency instead of time stamps to adapt traditional power analysis attacks to the new Differential Impedance Analysis (DIMA) and Correlation Impedance Analysis (CIMA). To perform such attacks, the adversary pauses the device's global clock signal, e.g., during the first round of a block cipher. This creates an idle state, allowing the adversary to measure the *S*-parameters through the device without any ongoing computations for an extended period of time. This procedure is similar to that of a SPSCA attack [23–26]. In contrast to the SPSCA attack, which can be susceptible to noise from dynamic power consumption during static snapshots without a halted clock, Impedance Side-Channel Analysis (IMSCA) is inherently resilient to time-varying measurements as long as the target intermediate values are stored in the circuit (e.g., in flip-flops). This robustness stems from the non-additive leakage of each individual intermediate value through the frequency domain, which significantly mitigates the need for stringent clock control in the proposed threat model. Consequently, the ability to pause the clock signal of the Device Under Test (DUT) is not a requirement for this type of attack as long as the targeted intermediate value stored in registers stays unchanged. A similar approach has been shown in [27] for an SPSCA attack. In the same work [13], the concept behind "virtual probes" was introduced, characterizing the chip using transmit carrier signals at different frequency bands. Hence, the attacker can simultaneously probe different registers using IMSCA. The authors also successfully attacked a masked Advanced Encryption Standard (AES) implementation and recovered all masked keys using a single-bit Template Impedance Attack (TIMA).

1.2 Our Contributions

The previously published attacks using IMSC [13, 28] and countermeasures [29] have relied on FPGAs. Given their ability to provide significant control over the placement and routing of circuit components, FPGAs introduces a level of flexibility that may influence the analysis results. In contrast, in this study, we employ a dedicated ASIC prototype chip fabricated with a commercial 28 nm CMOS technology to conduct the analyses in a more realistic environment.

We further present the results of an experimental evaluation of the effect of measurement factors (particularly supply voltage and ambient temperature) on the exploitability of IMSC and its practical verification. We try to answer whether the adversary can physically increase the amount of information leakage through the IMSC by controlling specific operating parameters. As a particular case study, we focus on a serial implementation of the PRESENT-80 block cipher [30] in a controlled environment for a different temperature-voltage combination (0 to 65°C, and 0.75 V to 1.35 V), which took roughly 60 days. In addition, we report on Moment-Correlating DPA (MCDPA) key-recovery attempts on the ASIC using the measured *S*-parameter.

In short, our experimental evaluations indicate a lower exploitable data dependency in the *S*-parameter at lower temperatures. Additionally, we show that both measurement factors can be combined to increase the information leakage through the IMSC and improve key-recovery attacks. We also show the challenges associated with noise during the measurement of the *S*-parameter in a climate chamber and how we can mitigate its effect on impedance measurements. Finally, the results of the SPSCA are presented, along with a comparison between SPSCA and IMSCA. Please note that an initial comparison between *dynamic* power analysis and IMSCA has already been presented in [28], where IMSCA outperformed dynamic power analysis. However, due to the similarity between the IMSCA presented in [13] and other static side-channel attacks in terms of attack requirements (e.g., halting the clock of the victim circuit), it is more interesting to compare IMSCA directly with SPSCA. The comparison focuses on the minimum required number of samples, the time needed to collect them, and a discussion of the ease of employing each method as a side channel.

2 Background

CMOS technology dominates the integrated circuit industry due to its multiple advantages. Naturally, dies, fabricated with CMOS transistors, are electrical circuits which can be modeled by their electrical impedance. Precise impedance measurements of the die can be carried out, even in the most advanced-node CMOS technologies. However, such precise impedance characterizations can inadvertently leak the state of a circuit on the chip's die [13]. In the following, we explain the fundamental concept of the *S*-parameter analysis, how it relates to impedance, and how information can be leaked through it. Since our analysis in this work focuses primarily on an implementation of the PRESENT-80 block cipher realized on a custom ASIC design as the case study, we also provide a concise overview of the block cipher and the design architecture of the underlying implementation.

2.1 Power Distribution Network and Sources of On-Die Impedance

The system's power delivery network (PDN) can be modeled by an RLC network (Fig. 1), connecting the voltage regulator module to every power-sinking component on the PCB, package, and die. PDN are optimized to deliver a stable, low-noise voltage supply to the power rails on the die [18, 31]. The impedance contribution of various

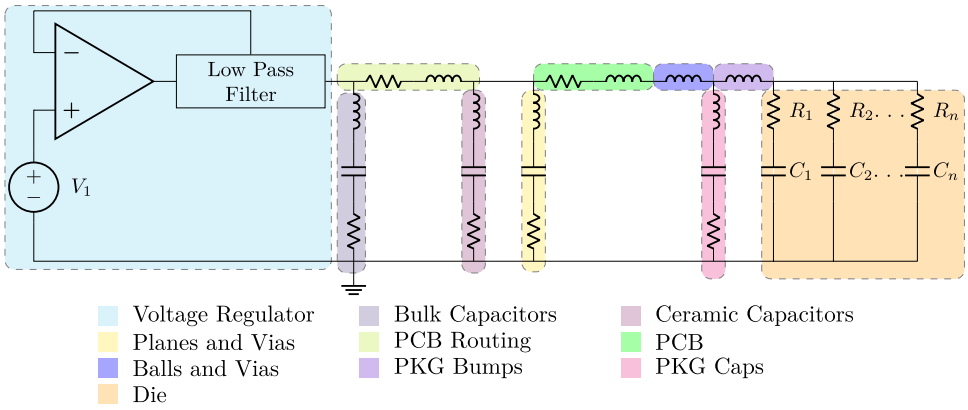


Fig. 1 Equivalent RLC circuit model of the power distribution network of the PCB and chip

components (on the PCB, package, and die) to the overall PDN impedance changes across different frequency bands. In [18], it is shown that while the equivalent impedance of the PDN at lower frequencies is dominated by the voltage regulator's and off-chip components' impedance, the impedance of on-chip components plays a more significant role at higher frequencies. While an ideal capacitor can be modeled as a short circuit at high frequencies, PDN's capacitors have parasitic inductances that cause a resonance at a particular frequency depending on the capacitance and inductance values. Consequently, the capacitor's impedance increases above its resonance frequency and should be modeled as an open circuit at very high frequencies, where its impedance increases significantly. Consequently, as the frequency goes up, all groups of capacitors, from the large to the small ones, become open circuits, and their impact on the PDN impedance decreases. Therefore, on-chip structures dominate the PDN impedance in higher frequency bands due to their smaller structural sizes and, hence, smaller parasitic inductance values. The authors of [4, 32, 33] showed how to detect PCB-level tamper events by characterizing PDN's impedance in the frequency domain. Alternatively, the authors of [13] have shown that tampering with the circuit's logic gates, routing, and placement alters the On-Die Capacitance (ODC). This alteration, in turn, affects the impedance from the PDN perspective, particularly in the specific frequency bands dominated by those ODC. The dominant frequency bands are determined by the chip's dimensions, placement, and the nature of the modifications. The relationship between impedance in these specific frequency bands, ODC, and On-Die Resistance (ODR) has also been demonstrated in [34, 35].

Figure 2 shows a CMOS inverter and its profile with the location of symbiotic ODCs. Some circuit elements, like well-junctions and interconnections within idle logic circuitry, can unintentionally contribute to ODC and, luckily, help to manage the power supply noise [36–38]. ODC is mainly formed by the metal power net, non-switching gate and p-n diode junction diffusion (between drain and substrate, drain and well, and well and substrate). The dominant contributor of ODC is the non-switching gate capacitance. ODR stems primarily from the resistance of the power network, the resistance of the transistor channel, transistor gate resistance, and resistance of contacts of n-well and p-substrate [39]. From now on, the acronym ODC will be referred to as C_{die} and the acronym ODR as R_{die} for the sake of simplicity. C_{die} can be divided into three primary groups.

- C_{m}, which is the capacitance associated with the on-die power/ground metallization grid network.
- C_{d}, which is associated with the p-n diode junctions.
- C_{g}, the gate capacitance, which is the main contributor to the total C_{die} [18].

All non-switching and powered-on circuits contribute to C_{g} in the chip's PDN. This phenomenon occurs because an operational transistor has a channel under the gate and contributes to C_{die}. In contrast, a powered-off transistor has an inactive channel and does not significantly contribute to the ODC. In the absence of device power, the effect of the gates' decoupling capacitance is unrecognizable. However, as the device is activated, the channels start forming, and consequently, C_{g} dominates C_{die}.

2.2 Information Leakage Through *S*-parameter

In this work, we use *S*-parameter to characterize the PDN's impedance versus frequency. This approach is utilized in some related state-of-the-art studies [4, 13, 40]. *S*-parameter are network parameters that describe the relationship between reflected and incident waves rather than the relationship between port voltages and currents [40]. They describe the behavior of electrical networks and represent the signal attenuation, reflection, and transmission ratio at each network port in response to the applied electromagnetic field [41]. In frequency domain analysis, waveforms are represented using sine waves. A sine wave can be fully characterized by frequency, amplitude, and phase. Therefore, we use the

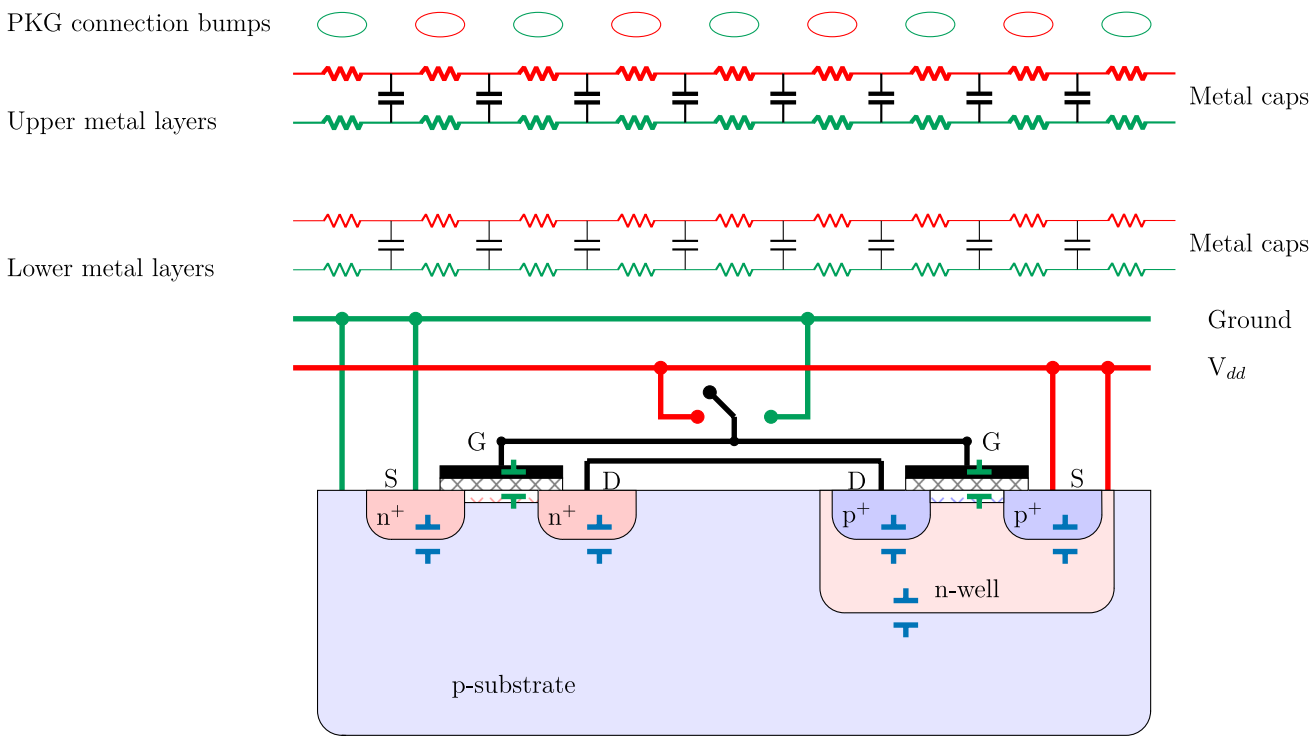


Fig. 2 Cross-sectional view of a CMOS inverter and on-Die capacitor locations. The black capacitors show C_{m}, the blue ones show C_{d}, and the capacitors shown in green corresponds C_{g}

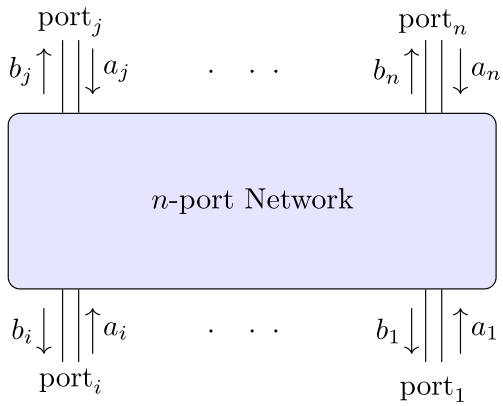


Fig. 3 Diagram of an n-port network representing S-parameter. a_i represents the incident wave applied at the -th port and b_i is the reflected wave from port_i

amplitude and phase responses in the frequency domain to accurately characterize the chip across the frequency range of interest, analyzing its behavior at discrete frequency points. The mathematical representation of the *S*-parameter for an *n*-port network is expressed as an *n*-by-*n* matrix (Fig. 3):

$$\boldsymbol{S} = \begin{bmatrix} S_{11} & S_{12} & \dots & S_{1n} \\ S_{21} & S_{22} & \dots & S_{2n} \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ S_{n1} & S_{n2} & \dots & S_{nn} \end{bmatrix}, \tag{1}$$

where S_{xy} can be defined as the ratio of the reflected wave at port x to the incident wave applied at port y while the incident wave at all other ports is set to zero.

Therefore, S_{11} represents the amount of reflected power from port 1 if the Electromagnetic (EM) field is applied at port 1. The EM wave is represented by an sinusoidal wave (sine wave), which can easily be characterized by its frequency, phase, and amplitude. One option to measure S_{11} is to employ a Vector Network Analyzer (VNA). A VNA is a precision instrument that measures the electrical properties of components at high frequencies. It sends a known sine wave through a device and analyzes the reflected and transmitted signals. A VNA can determine the device characteristics, such as *S*-parameter, impedance, gain, and phase shift, by comparing the given and reflected signals. To record the reflection coefficient of the circuit from the PDN ports (including the DUT's PDN) at discrete frequency points, the VNA transmits the sine wave with different frequencies in the desired/selected range. The impedance profile is a straightforward derivation from the reflection coefficient. In this study, S_{11} is related to the equivalent impedance of the circuit from the PDN ports.

The relationship between the equivalent impedance of the DUT from its PDN point of view and its reflection coefficient from the same perspective (S_{11}) is expressed in Eq. 2 [41].

$$Z_{\mathrm{DUT}} = Z_0 \frac{1 + S_{11}}{1 - S_{11}}, \tag{2}$$

where Z_0 represents the reference impedance of the VNA which is equal to 50Ω, Z_{DUT} correspond to the impedance obtained from S_{11}, and S_{11} is the reflection coefficient. It should be mentioned that Z_{DUT} is the impedance of the whole PDN network, including the impedance of the Voltage Regulator Module (VRM) (Z_{VRM}), Bulk and ceramic capacitors on the PCB ($Z_{\mathrm{PCB}_{\mathrm{caps}}}$), PCB planes and vias ($Z_{\mathrm{PCB}_{\mathrm{rout}}}$), on-chip PDN ($Z_{\mathrm{chip}_{\mathrm{PDN}}}$), and the cryptographic core (Z_{core}).

$$Z_{\mathrm{DUT}} = f(Z_{\mathrm{VRM}}, Z_{\mathrm{PCB}_{\mathrm{caps}}}, Z_{\mathrm{PCB}_{\mathrm{rout}}}, Z_{\mathrm{chip}_{\mathrm{PDN}}}, Z_{\mathrm{core}}) \tag{3}$$

As the authors of the [13] have shown, each part of Z_{DUT} dominates a specific range of the frequency span, and in high frequencies, Z_{DUT} is dominated by $Z_{\mathrm{chip}_{\mathrm{PDN}}}$ and Z_{core} which are data dependent.

We analyze the propagation of the voltage wave injected by the VNA into the chip using an ideal transmission line model to understand how the chip's behavior affects the injected signal. This fundamental model provides a basis for understanding more complex circuit behavior and clarifies the underlying principles of our methodology. Figure 4 illustrates an ideal transmission line model in which the characteristic impedance (η) and the propagation constant (γ) of medium 2 undergo changes. For the sake of simplicity, medium 1 and medium 3 are assumed to be lossless, resulting in a characteristic impedance of η_0 and a propagation constant of $\gamma_0 = j\beta_0$. Medium 2 is considered a non-magnetic material ($\mu_r = 1$) with a relative permittivity of ε_r. Given that ε_0 and μ_0 represent the permittivity and permeability of free space, respectively, and β_0 denotes the free-space wave number, we can express β_0 as $2\pi f\sqrt{\varepsilon_0/\mu_0}$ and reformulate the propagation constant of the second medium as $\gamma = j\beta_0\sqrt{\varepsilon_r}$. Furthermore, using $\eta_0 = \sqrt{\mu_0/\varepsilon_0}$, the characteristic impedance of medium 2 can be rewritten as $\eta = \sqrt{1/\eta_0\varepsilon_r}$. The VNA applies a voltage wave with a known amplitude of V_1^+ in medium 1, and the reflected voltage wave has an amplitude of V_1^-. Multiple reflections and transmissions occur within the transmission lines upon the injection of V_1^+. $V_1(z) = V_1^+ e^{-j\beta_0 z} + V_1^- e^{+j\beta_0 z}$, $V_2(z) = V_2^+ e^{-\gamma z} + V_2^- e^{+\gamma z}$,and$V_3(z) = V_3^+ e^{-j\beta_0 z}$,where V_i^+ and $V_i^-(i = 1, 2, 3)$ denote the forward and backward voltage waves in medium i. For simplicity, it is assumed that there is no backward voltage wave in medium 3.

V_1^+ is a known parameter injected by the VNA, while V_1^-, V_2^+, V_2^-, and V_3^+ are unknowns. All four unknowns can be determined by applying the boundary conditions for voltage wave components at the interfaces between media. Our objective is to derive the reflection coefficient S_{11} in medium 1, which is given by

$$S_{11}(f, \varepsilon_r, L) = \frac{V_1^-}{V_1^+} = \frac{(\eta^2 - \eta_0^2)(1 - e^{+2j\gamma L})}{(\eta_0 + \eta)^2 - (\eta - \eta_0)^2 e^{+2j\gamma L}}, \tag{4}$$

where L represents the length of the path traveled by the injected voltage wave. Equation 4 shows that the reflection coefficient is influenced by three factors: the frequency band of interest, the sample's relative permittivity, and the length of the propagation path. Moreover, the frequency dependence of S_{11} has an additional implication: since frequency and wavelength are inversely related, more minor changes in the chip's configuration can be detected in higher frequencies. When incident waves from the VNA interact with registers having different placement and routing configurations, variations in the parameters of Eq. 4 lead to corresponding changes in the S_{11} profile at specific frequencies. For instance, altering the placement and routing of the circuit changes L, which in turn affects the reflection response for different configurations.

2.3 Present Block Cipher

PRESENT is a block cipher that operates on a 64-bit block and has been designed specifically for resource-constrained

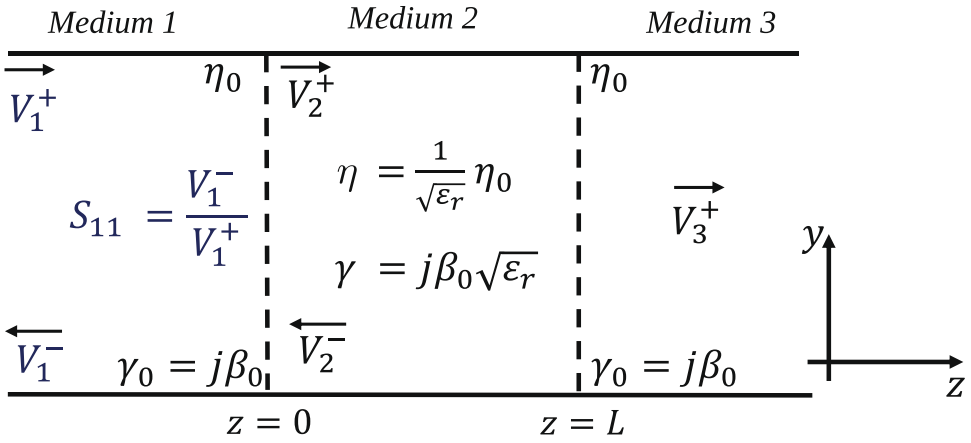


Fig. 4 The simplified model of a transmission line illustrating the incidence of normal uniform plane waves on different media, where the characteristic impedance of medium 2 differs from that of medium 1 and medium 3. (Taken from [13])

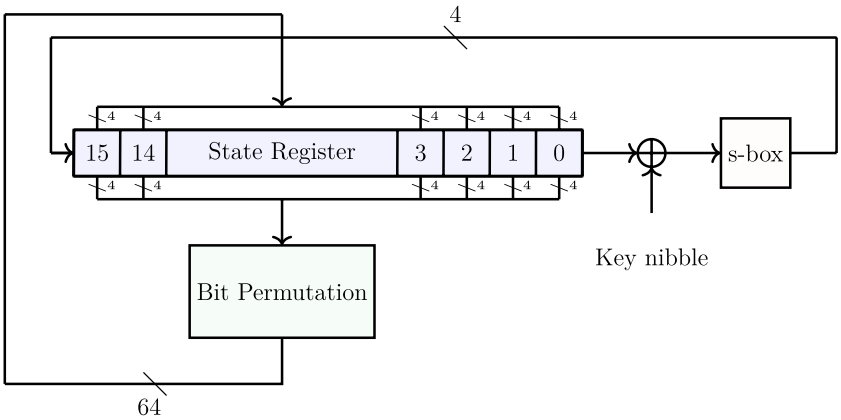


Fig. 5 Block diagram of a hardware implementation of the nibble-serial PRESENT block cipher. The key schedule is not shown

devices, including Radio Frequency Identification (RFID) tags, Internet of Things (IoT) devices, and wireless sensor networks [30]. The cipher supports key lengths of 80 bits and 128 bits, and its design is based on a Substitution-Permutation Network (SPN), which consists of 31 rounds. The cipher employs a lightweight 4-bit S-box to ensure both security and implementation efficiency. Given its minimal hardware footprint, PRESENT is an optimal choice for deployment in ultra-constrained environments with limited computing resources. Figure 5 depicts a nibble-serial implementation of the cipher, which has been optimized for area efficiency in accordance with the implementation presented in [42]. This implementation processes one nibble per clock cycle and has a minimal area footprint, rendering it one of the most resource-efficient block ciphers.

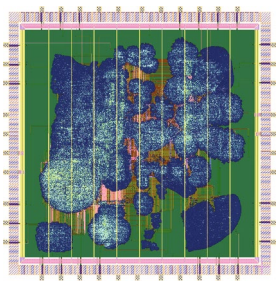


Fig. 6 Layout of our fabricated ASIC prototype

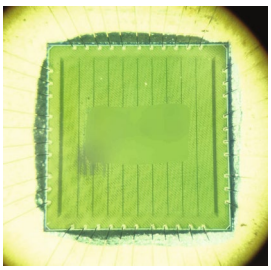


Fig. 7 Microscopic photo of Our fabricated ASIC prototype

3 Experimental Setup

This section provides a description of the target device, the measurement setup, and the procedure used for the experimental analyses.

3.1 Physical Target Device

In our experimental analysis, we used a 28 nm CMOS ASIC prototype developed as a dedicated test chip for our research. The microscopic photo and the layout of the fabricated and bonded die are depicted in Figs. 6 and 7. The chip's physical dimensions are 1380 µm × 1380 µm. The DUT can operate at frequencies up to 100 MHz, even under challenging conditions. Its I/O and core nominal supply voltages are 1.8 V and 0.9 V, respectively. The standard-cell area in the center of the die measures 981 µm × 981 µm and contains 11 cryptographic co-processors. These 11 cores are not identical because they differ in the countermeasures employed to prevent key extraction via various side-channel analyses. The implemented countermeasures do not target impedance-based analysis, and therefore the protected cores fall outside the scope of this study. The levels of protection range from an unprotected implementation to designs incorporating balancing and provably secure masking. However, this study focuses only on one of these cores, namely the raw and unprotected nibble-serial PRESENT implementation described in Section 2.3 and shown in Fig. 5.

3.2 Measurement Setup

Figures 8 and 9 illustrate the measurement diagram and the measurement setup used in this study. An evaluation board comprising two distinct parts is designed and constructed, as shown in Fig. 10. A 48-pin Dual In-line Package (DIP) socket is located on the left-hand side, where it is employed for the purpose of connecting a Digilent Cmod A7 FPGA board. The ASIC chip hosted in a 44-pin Plastic Leaded Chip Carrier (PLCC) socket is placed on the right-hand side. The FPGA acts as an interface between the ASIC, Source Measure Unit (SMU), VNA and the PC. The FPGA

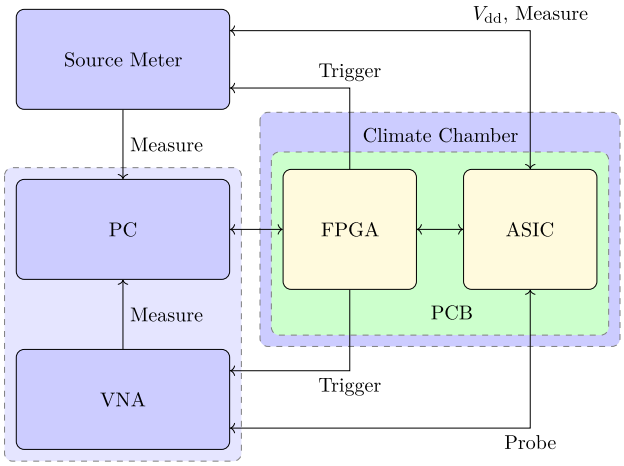


Fig. 8 Diagram of the measurement setup used in our practical experiments

serves to regulate the measurement process, and communicates with the measurement setup using a trigger signal from an I/O port of the FPGA. A Keithley 2450 SMU is simultaneously used as a power supply and high-precision current measurement instrument. One of the objectives of this study is to examine the impact of the temperature on the leakage through the IMSC. As commonly done in the state of the arts (e.g., in areas such as SPSCA [26, 43, 44]), the evaluation board, including the ASIC chip, is placed in a climatic chamber, particularly a C series climatic test cabinet from CTS company, to ensure precise control of the environmental temperature. To measure the S_{11} parameter, a Keysight ENA E5080B VNA was utilized, which is capable of conducting RF/microwave scattering measurements, and operates within a frequency bandwidth of 9 kHz to 14 GHz. While the sine wave is supplied to the PDN with the supply voltage connected, it is necessary to filter out the DC voltage on the VNA ports. Accordingly, the VNA ports utilized in these investigations are equipped with internal capacitors in series to prevent the flow of DC current. The communication between the PC and the

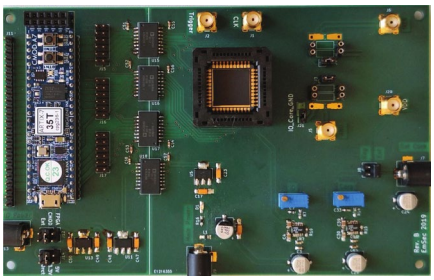


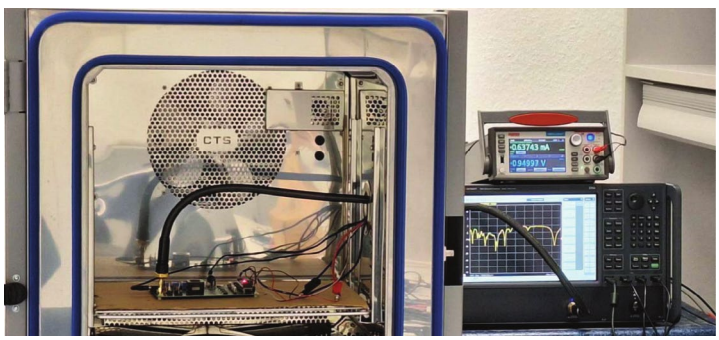
Fig. 10 Dedicated PCB developed for our measurement setup

FPGA board is facilitated via a Universal Asynchronous Receiver-Transmitter (UART) port.

3.3 Measurement Procedure

The FPGA is responsible for transmitting the plaintext and key to the ASIC for every encryption. Consequently, in the event that the measurement necessitates a random plaintext, the FPGA is responsible for generating this internally and transmitting it to the ASIC. Subsequently, the clock of the ASIC chip is halted by the FPGA during the first round of each PRESENT encryption at the specified clock cycle. Immediately after holding the chip clock, the FPGA generates two trigger pulses, one for the SMU and the other for the VNA. To prevent any interference between the static leakage and S_{11} measurements, each SMU and VNA is scheduled to perform its respective measurement within a designated time interval. Consequently, upon receiving the trigger pulse, the SMU waits approximately 20 ms to allow the electrical quantities of the ASIC to stabilize to their DC values. Subsequently, the SMU acquires the supplied current value and transmits it to the PC as the corresponding static leakage. Afterward, approximately 80 ms after the trigger pulse is received by the VNA, it starts measuring S_{11}, a vector of complex numbers corresponding to specific frequencies. The data is then transmitted to the PC, where the phase and amplitude of each element of the S_{11} vector

Fig. 9 Measurement setup used in our practical experiments



are calculated using these complex numbers. While awaiting the next trigger pulse, FPGA releases the chip clock pulse, allowing it to complete the encryption process (to verify the calculation's correctness) before starting a new encryption.

During the measurements, we observed that using the climate chamber introduces considerable noise into the S_{11} measurements over a wide frequency range. We have determined that the rotary fan of the climate chamber is the source of this noise. Specifically, we found out that the rotary fan introduces electromagnetic interference, likely through radiated emissions from the fan motor, which corrupts the S_{11} measurements. Therefore, to eliminate the noise, it became necessary to pause the climate chamber during the measurements of the *S*-parameter. As a result, we changed the measurement procedure as follows. Initially, the climate chamber works ordinarily to reach the desired temperature. Subsequently, the climate chamber is placed in a state of suspension. Afterward, a 10-second wait should be ensured to allow for the cessation of the rotary fan. After completing a series of measurements, the data collection process is halted, and the climate is turned back on, allowing it to resume the temperature-control functions for a designated period of 90 seconds. This procedure is then continued to collect the necessary number of measurements. The number of measurements that can be collected in each round depends on the desired temperature. When the temperature difference between the ambient environment (room temperature) and the inside of the climate chamber is increased, the rate of temperature change during the measurement phase also increases. This can lead to complications such as misalignment of the collected measurements, which we address in the next section. Figures 11 and 12 highlight the effect of noise originating from the rotary fan of the climate chamber and the effectiveness of our measurement procedure to eliminate it.

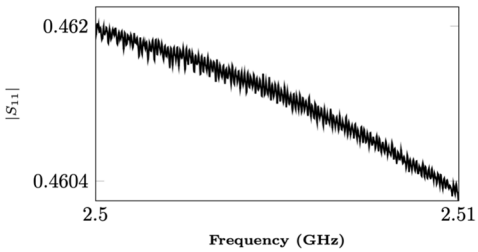


Fig. 11 The effect of climate chamber's fan on S_{11} measurement at 25° C

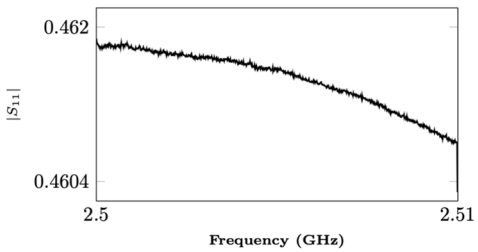


Fig. 12 The effectiveness of our measurement procedure on S_{11} measurement at 25° C

4 Analyses

This section presents the results of analyses conducted using data collected over approximately two months of continuous data acquisition. Following this, a comparative assessment of SPSCA and IMSCA is provided, focusing on key factors such as leakage assessment results, time required for the assessment, ease of use, measurement cost, and overall effectiveness of each side channel.

4.1 S_{11} Measurement

As it has been shown earlier in Fig. 1 from the VNA point of view (from the outside of the chip looking inside), the system's PDN can be modeled as a series RLC network. In this model, L is the inductance of the package, which models the package bumps, R is the R_{die}, and C is the C_{die}. However, from the on-die circuit point of view (from the inside of the chip looking outside), the circuit is a series RC in parallel with a series RL [34] (Figs. 13 and 14). However, we are only considering the RLC series model in this study, as it can be relatively easily measured on board non-invasively. Simulated frequency response of a series RLC shown in Fig. 15 depicts that the capacitance produces a pole and always manifests as a -20 dB per decade slope, and the inductance always produces zero and manifests as a $+20$ db per decade slope in a log-log scale [34]. At the same time, the resistance can be identified as the bottoming of the response in Fig. 15. However, in a real integrated circuit such as an FPGA or an ASIC, a multitude of components contribute to the frequency response, making the situation more complex than that of a single RLC circuit. For instance, Figs. 16 and 17 depict the S_{11} curve across the frequency range of 1-4 GHz for our custom ASIC, where the

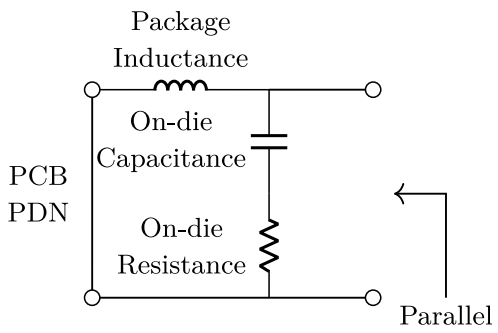


Fig. 13 Modeling the system's PDN as an RLC circuit from the on-die circuit perspective

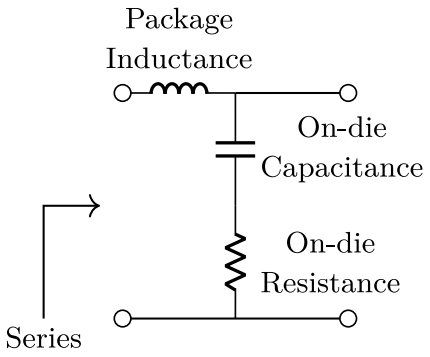


Fig. 14 Modeling the system's PDN as an RLC circuit from the VNA perspective

effect of substrate resonances through the local poles/zeros are clearly observable.

As depicted in Fig. 16, the amplitude of the S_{11} parameter exhibits local minimums and maximums, which can be attributed to the presence of zeros and poles in its transfer function. At the same time, the influence of zeros on the phase of S_{11} is observable in Fig. 17. The phase exhibits a pronounced increase due to the presence of zeros, whereas a steep decline in phase indicates the impact of a pole. However, analyzing all poles and zeros in a complex circuit is impractical due to the complexity. In fact, by looking at such a large-scale S_{11} graph, we can only identify major poles and zeros. Therefore, if we look at the *S*-parameter on a high-resolution graph, each slope of the *S*-parameter graph contains many small peaks and dips related to the tiny circuit elements, such as parasitic capacitors. In this study, our focus is on understanding the effect of environmental parameters on these small elements, which probably can affect the data dependency of S_{11} on these environmental parameters. To detect hidden data dependencies that are not visually apparent, we deploy the t-test as a leakage assessment tool.

Before evaluating the circuit through a t-test analysis, we look at the effect of the temperature and supply voltage on the amplitude and phase of S_{11}. For better visibility,

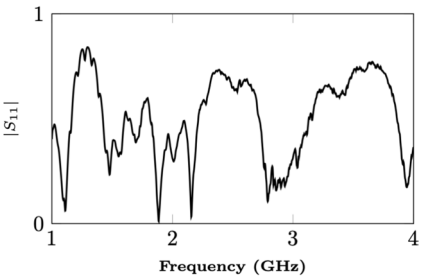


Fig. 16 A general $|S_{11}|$ representation of our ASIC over a 3 GHz frequency bandwidth

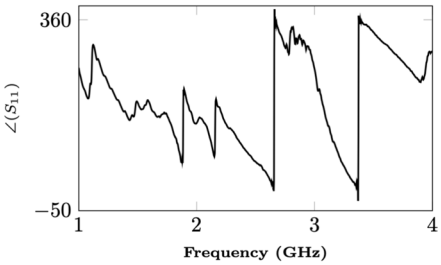
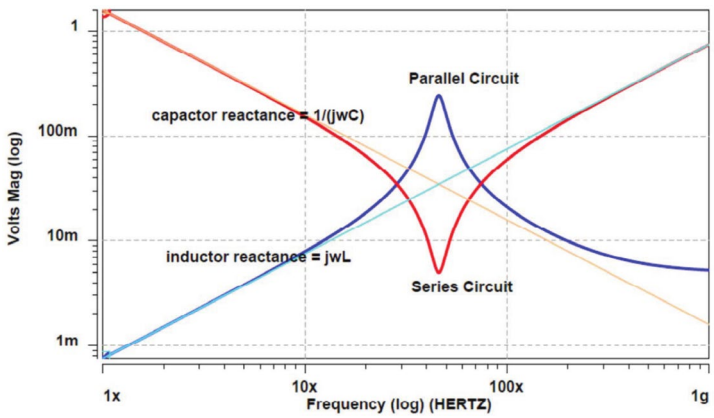


Fig. 17 A general $\angle(S_{11})$ representation of our ASIC over a 3 GHz frequency bandwidth

we limit the S_{11} graph around its first dip in Fig. 16 for the frequency range 1.11 GHz-1.12 GHz. Figures 18 and 19 depict the effect of variation in the temperature on S_{11} for the temperature range of 15 °C-65 °C with 10 °C steps while the supply voltage is fixed to 1.2 V. As illustrated in Figs. 18 and 19, an increase in temperature results in a shift of the amplitude and phase of the S_{11} curve to the left along the frequency axis. This indicates that the system's poles and zeros migrate towards lower frequencies with increasing temperatures. This phenomenon is consistent with the findings of [45], and we can translate this movement as an increment in the size of the parasitic elements of the circuit. The other detectable effect of temperature on the *S*-parameter graph is that the minimum of $|S_{11}|$ moves slightly upward. The dips in the $|S_{11}|$ plot are associated with the circuit's resonance frequency, where zeros and poles neutralize each other at

Fig. 15 Simulated frequency response of sample parallel and series RLC circuits



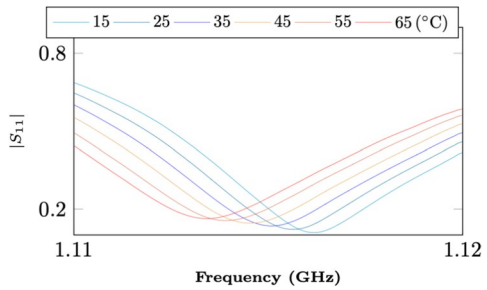


Fig. 18 The effect of variation in temperature on $|S_{11}|$

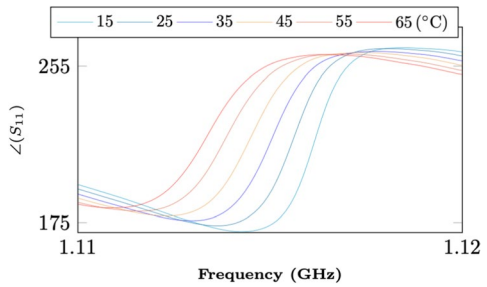


Fig. 19 The effect of variation in temperature on $\angle(S_{11})$

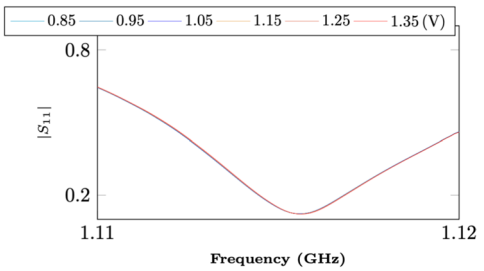


Fig. 20 The effect of variation of supply voltage on $|S_{11}|$, measured at 25 °C

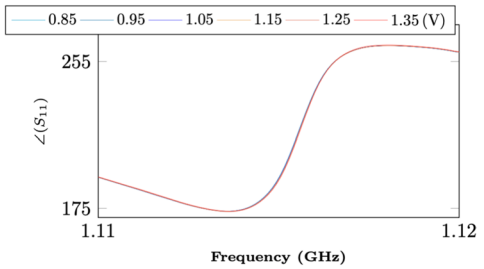


Fig. 21 The effect of variation of supply voltage on $\angle(S_{11})$, measured at 25 °C

that frequency. Therefore, if we consider the PDN model as a series RLC, the circuit's resistor determines the impedance amplitude in the resonance frequency. Based on this assumption and with reference to Eq. 2, it can be inferred that an upward shift of S_{11} along the vertical axis implies an increase in the resistor value.

Unlike the analog circuits, in which almost all Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET transistors work in the saturation region, in a digital circuit, every single transistor, based on its gate voltage, is in cut-off or deep triode region, i.e., $V_{\mathrm{DS}} \ll 2(V_{\mathrm{GS}} - V_{\mathrm{th}})$. In the deep triode region, the transistor operates as a resistor [46]:

$$R_{\mathrm{on}} = \frac{1}{\mu C_{\mathrm{ox}} \frac{W}{L}(V_{\mathrm{GS}} - V_{\mathrm{th}})}, \tag{5}$$

where μ is the carrier mobility (electron mobility for NMOS, hole mobility for PMOS), C_{ox} is oxide capacitance per unit area, W is the width of the MOSFET channel, L is the length of the MOSFET channel, V_{GS} is Gate-to-source voltage, and V_{th} is Threshold voltage.

Referring to Eq. 5, a reduction in the threshold voltage of a MOSFET transistor resulting from an elevation in temperature [47–49], should be accompanied by a corresponding decline in the R_{on} value for a MOSFET transistor operating in the triode region. However, as previously outlined in [50], the electron-phonon scattering rate is reduced due to an increase in temperature, which in turn leads to an increase in R_{on}. It appears that the effect of the second factor is more substantial. Therefore, the overall equivalent resistor of the circuit is increased due to an increase in the temperature [51]. Additionally, the *S*-parameter measurement results, as illustrated in Figs. 18 and 19, indicate that maintaining a constant temperature throughout the measurement process of the *S*-parameter is essential for accurate alignment of the collected measurements.

The effect of V_{gs} on parasitic capacitances is complex and depends on the effective frequency [52–55]. For low frequencies, the value of C_{g} in the inversion region ($V_{\mathrm{gs}} > V_{\mathrm{th}}$) slightly increases as V_{gs} increases, while for high frequencies, C_{g} in the inversion region slightly decreases as V_{gs} increases [47]. However, based on the findings in [45], the supply voltage (which consequently affects the gate voltage) does not significantly change the gate capacitance of a CMOS transistor. Additionally, according to Eq. 5, an increase in V_{gs} slightly decreases R_{on}. Figures 20 and 21 illustrate the impact of varying the supply voltage on S_{11} at 25 °C for the supply voltage range of 0.75 V-1.35 V in 0.1 V ascending steps. As shown in Figs. 20 and 21, changing the supply voltage does not have a significant effect on S_{11} within the specified supply voltage range.

4.2 Leakage Assessment

To investigate different working conditions, we needed to repeat the measurement procedure for each environmental parameter (i.e., supply voltage and temperature) to collect 40,000 samples. The supply voltage was increased in steps of 0.1 V (7 different supply voltages) within the range of 0.75 V to 1.35 V, to investigate the effect of supply voltage on IMSCA. For each supply voltage, we have also studied 5 different temperatures: 0° C, 15° C, 25° C, 35° C, and 65° C to evaluate the effect of temperature on IMSCA. Given the number of required measurements and the time-consuming nature of low-noise S_{11} measurements, we decided to focus on a limited frequency band. As a result, we collected 35 different voltage-temperature combinations, which took about two months to complete. To determine the appropriate frequency range, the S_{11} parameter was measured across a broad spectrum, and the desired frequency range was subsequently selected based on the results of the leakage assessment. To conduct a leakage assessment using the t-test and compare the data dependency of the static and impedance side channels of the PRESENT core under different conditions, measurements have been taken for two different fixed inputs: one with all bits set to 1 and the other with all bits set to 0 (fixed-vs-fixed) in a randomly interleaved manner [56]. Based on the previous measurements (see Figs. 18 and 19), it has been observed that temperature variation results in shifts in both the amplitude and phase of S_{11} over the horizontal and vertical axes. Consequently, focusing on a very narrow frequency window may result in erroneous analysis, as a fixed frequency range displays different segments of the S_{11} curve for different conditions, i.e., different temperatures. Therefore, selecting an appropriate frequency range is crucial. Figure 22 depicts the chosen window, in the frequency range of 1.1125 GHz-1.1250 GHz with 50 kHz steps, which exhibits a discernible top in the middle of the selected frequency range. As indicated in Fig. 22, for temperature variation from 0° C to 65° C, the top remains within the selected range, ensuring that the crucial elements of the measurements are not overlooked. To minimize the noise effect and facilitate the comparison, the t-value for different scenarios was calculated as the average of multiple points around the dip of the t-test curve. The results of both voltage and temperature variations are combined and presented in Fig. 23.

In preliminary runs, we observed that temperature variations shift the S-parameter features along the frequency axis and slightly in amplitude Figs. 18 and 19, so that a fixed narrow frequency window can correspond to different portions of the response at different temperatures Fig. 22. This non-stationarity inflates the variance of a random class and reduces the power of fixed-vs-random Welch t-tests unless substantially more traces are acquired. Section 4.3.3 indicates that S-parameter acquisition is time-consuming; we therefore used fixed-vs-fixed because it provides higher power against mean-shift leakage with fewer traces and is less sensitive to the random-class variance inflation that fixed-vs-random suffers from in our setup.

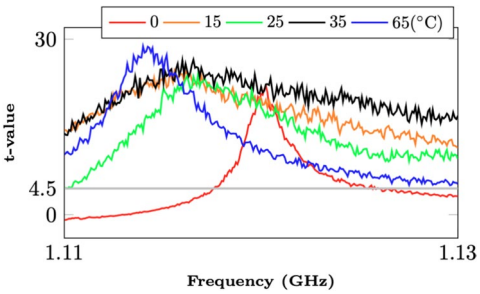


Fig. 22 The shift of the data dependency of the measured *S*-parameter across the frequency axis under different temperatures

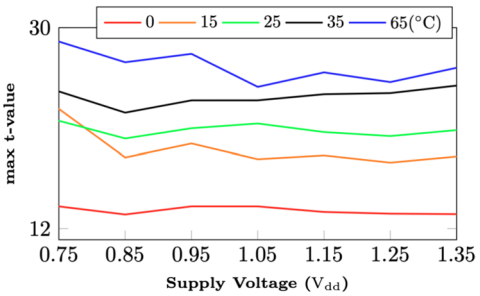


Fig. 23 The effect of variation in both supply voltage and temperature on the t-test results

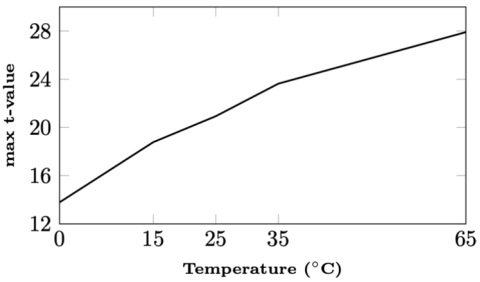


Fig. 24 The effect of variation in temperature on t-test results

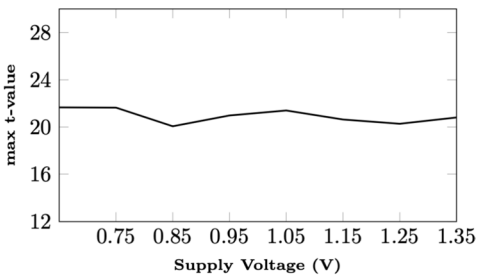


Fig. 25 The effect of variation in supply voltage on t-test results

Figures 24 and 25 depict the t-test results for various temperatures and supply voltages. As shown in Fig. 24, the

t-value increases from 13 to 28 as the temperature rises from 0° C to 65° C. It appears that the results are consistent with those reported in [57], which illustrates that all leakage effects that are based on solid-state physics, such as sub-threshold leakage and diode currents, show extreme thermal dependencies. Similar results can also be found in [26], which concludes that an increase in temperature can reduce the number of required measurements to exploit the leakages through the SPSCA. Notably, the backscattering parameter exhibits a pronounced thermal dependency, with a higher sensitivity to temperature than to supply voltage. On the other hand, the device's impedance is primarily determined by the circuit's structure and the materials used to construct the circuit. The structure of the circuit and also the electrical characteristics of the circuit materials are not voltage-dependent. Accordingly, as we expected, and has been shown in Section 4.1, the impedance characteristic of the circuit is not affected noticeably by the slight variation in supply voltage (i.e., variation of the supply voltage in the allowed range for our ASIC). Consequently, as illustrated in Fig. 25, the effect of the supply voltage is minimal, with the t-value remaining in the same range for supply voltages in the range of 0.75 V to 1.35 V.

4.3 SPSCA versus IMSCA

Here, we provide a comparative analysis of SPSCA and IMSCA.

4.3.1 Leakage Assesment

To compare the leakage of the PRESENT core via static power and impedance as two distinct side channels, we followed the same procedure explained above and performed a similar fix-vs-fix t-test. The influence of environmental factors on static power has been thoroughly explored in [26]. Therefore, we emphasize that we did not repeat that study here. However, we followed the same procedure to collect static power measurements. We collected the measurements in five specific environmental conditions, which contain the measurements related to four boundary operating conditions (0° C, 0.75 V), (0° C, 1.35 V), (65° C, 0.75 V), and (65° C, 1.35 V) and one nominal operating condition (25° C, 0.95 V). The number of recorded measurements in all cases is the same across the static power and S_{11} measurements, i.e., 40,000. In the case of static power measurements, each measurement results in a singular value, which is the amount of static power associated to the input applied to the PRESENT core under the selected environmental condition. Therefore, we have a set of 40,000 scalar measurements for each static power measurement under each specific operating condition. On the other hand, in the case of impedance measurements, a vector of S_{11} parameters under the same condition has been collected. More precisely, the VNA scanned the frequency range of 1.1125 GHz-1.1250 GHz with 50 kHz steps. Therefore, we have a set of measurements of 40,000 vectors, and each vector contains 251 complex numbers for each *S*-parameter measurement for every specific operating condition.

We performed a leakage assessment using the phase data, as described above. Therefore, the leakage assessment based on S_{11} measurements produced a vector of 251 elements, each of which is related to a specific frequency in the aforementioned frequency range. To show the t-test results as a plot of the number of measurements to be comparable to the static power results, we calculated the average of the 20 largest t-values across all 251 frequencies. The corresponding results are shown in Figs. 26, 27, 28, 29, 30, 31, 32, 33, 34 and 35. We first should highlight that our static power results align entirely with the findings reported in [26]. Further, in the absence of IMSCA key recovery, comparing t-values for static power and *S*-parameter under the same environmental conditions, it can be seen that static power noticeably outperforms the *S*-parameter. Additionally, the minimum number of required samples to have a t-value over the threshold

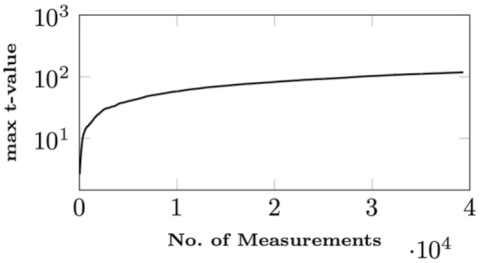


Fig. 26 The t-test results for static side channel at 0 °C, 0.75 V

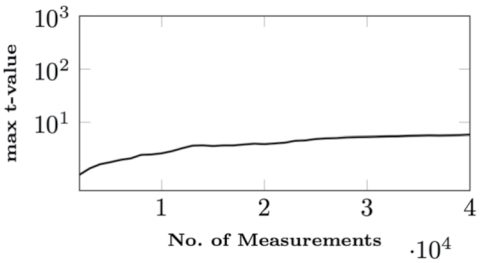


Fig. 27 The t-test results for impedance side channel at 0 °C, 0.75 V

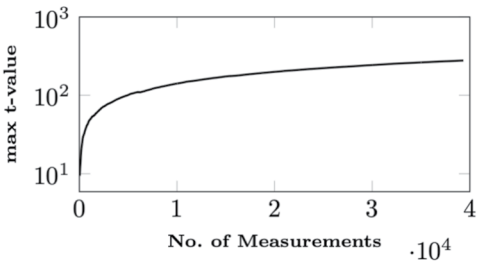


Fig. 28 The t-test results for static side channel at 0 °C, 1.35 V

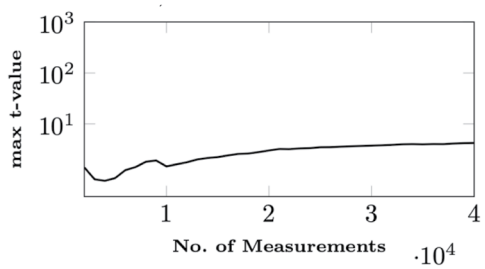


Fig. 29 The t-test results for impedance side channel at 0 °C, 1.35 V

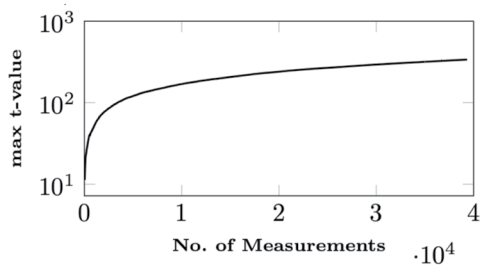


Fig. 30 The t-test results for static side channel at 25 °C, 0.95 V

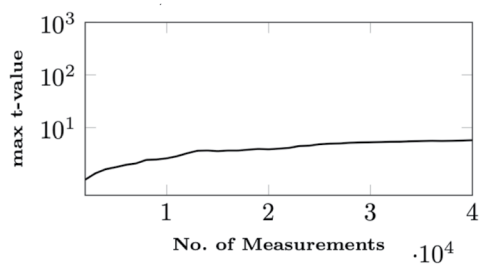


Fig. 31 The t-test results for impedance side channel at 25 °C, 0.95 V

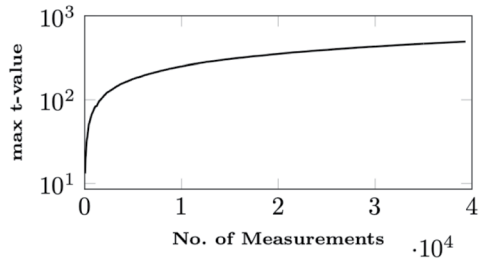


Fig. 32 The t-test results for static side channel at 65 °C, 0.75 V

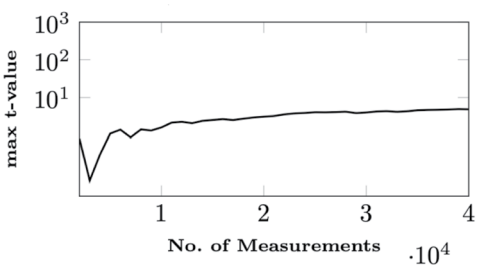


Fig. 33 The t-test results for impedance side channel at 65 °C, 0.75 V

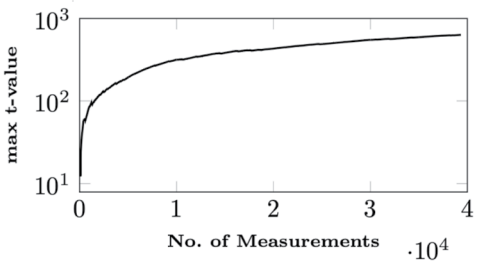


Fig. 34 The t-test results for static side channel at 65 °C, 1.35 V

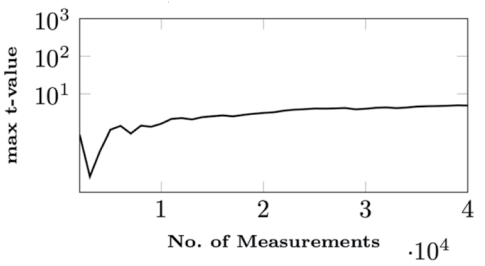


Fig. 35 The t-test results for impedance side channel at 65 °C, 1.35 V

Table 1 Comparison between SPSCA and IMSCA based on the minimum number of required measurements for a t-value over the threshold (4.5), t-value using 40,000 measurements, the required time to record 40,000 measurements, and the sensitivity to noise. Environmental condition: 25° C, 0.95 V

	min. no. of measurements	max.t-value	measurement time	noise sensitivity
SPSCA	4	596	1 hour	high
IMSCA	1200	20	13 hours	very high

(>4.5) in the case of static power is considerably lower than that for *S*-parameter. These results reflect that, under identical lab constraints, SPSCA currently offers far higher practical attack efficiency than IMSCA on our ASIC, despite IMSCA showing clear temperature-dependent leakage The comparison results at an ordinary condition (25 °C, 0.95 V) are summarized in Table 1.

4.3.2 Practical Exploitability and Attempted MCDPA Attacks

To assess whether the observed IMSCA leakage can be turned into a concrete key-recovery attack on our ASIC, we implemented a MCDPA[5]. MCDPA is a collision-based technique that does not rely on an explicit hypothetical leakage model. Instead, it exploits collisions between intermediate values processed in (ideally) time-multiplexed hardware units and, by comparing the side-channel characteristics of two S-box evaluations, can recover the linear difference between the corresponding key portions (e.g., key nibbles). This attack strategy is well-suited to the nibble-serial PRESENT architecture of our DUT (see Section 2.3), in which a single S-box instance is reused to process all state nibbles. For the practical experiments, we collected impedance measurements for 40,000 uniformly random plaintexts under the voltage-temperature combinations that yielded the highest t-values in our leakage assessment. Following the approach adopted in related static-leakage works [23–26, 43, 44], we applied MCDPA to the complex S11 samples within the selected window (1.1125–1.1250 GHz, 50 kHz step size). However, a successful key recovery was not feasible within this work. The main limitation is the acquisition time for frequency-domain scanning. Moreover, exploitable leakage may occur outside the selected window. Achieving a realistic chance of success would therefore require many more traces over a much wider frequency band. Covering a broader span (e.g., comparable to Figs. 16 and 17 at 50 kHz resolution with more than 40,000 traces per point would extend the measurement time to several months for each environmental condition, which is well beyond the scope of the present study. As a consequence, we treat the MCDPA results as negative evidence regarding the practical exploitability of IMSCA, on this ASIC under realistic acquisition constraints, and focus our discussion on leakage detectability and comparative analysis against SPSCA.

4.3.3 Required Time

If we compare the required time to record 40,000 samples, around 1 hour would suffice to collect those number of static power measurements, while this was around 13 hours of continuous measurement in the case of *S*-parameter at 25° C, which was the same as the room temperature and the number of pauses in the measurements was low (to eliminate the effect of climate chamber's fan, see Section 4.1). However, as we mentioned earlier, when the temperature difference between the ambient (room temperature) and the inside of the climate chamber increases, the rate of temperature change during the measurement phase also increases. Therefore, to have a clean set of measurements, we had to decrease the number of measurements in each round, resulting in more pauses (each 100 seconds) between the subsequent measurements. For example, to collect 40,000 samples at 65 °C, we had to limit the number of measurements in each set to 40, which introduced 1000 pauses. Such a large number of pauses added a significant overhead (almost 11 hours) to our measurements.

4.3.4 Ease of Use

In terms of the simplicity of using these side channels, performing a successful SCA attack requires an exploitable leakage. As it has been explored in several studies, SPSCA can quickly reveal confidential information with a few measurements [23–26, 44]. However, although a successful attack using IMSCA on an FPGA-based implementation has been reported in [13], in a more realistic scenario, like targeting a non-profiled ASIC, the frequency range that includes exploitable information is unknown. Therefore, the attacker has to scan a wide frequency range[2] with high accuracy and perform the attack over the whole frequency components, which is time-consuming. In a realistic scenario, our device takes almost 180 seconds to scan a frequency range of 1 GHz-10 GHz with a step size of 10 kHz and a Intermediate Frequency (IF) filter bandwidth of 500 Hz.

4.3.5 Measurement Cost Comparison

Another parameter we can consider is the measurement cost in each scenario. To measure static power, we need a source meter. The SMU we utilized costs around € 8 k, while the price for the VNA is comparatively higher and depends on the characteristics of the device[3]. Additionally, in this study, we did not consider high-frequency considerations in PCB design. However, if we do consider such factors, the cost of PCB design for IMSCA could be higher than that for SPSCA. Furthermore, designing a high-frequency PCB requires more effort than designing a standard PCB.

[2] The possible frequency range is limited by the specification of the employed VNA.

[3] The VNA we employed costs around € 80 k.

4.3.6 Effectiveness Against Protected Implementations

The reported results in this work are acquired from an unprotected cryptographic implementation. It should be noted that IMSCA could still pose a more powerful threat to protected implementations. As shown in [13], based on the content bits and physical location of registers on the chip, S_{11} is changed at various frequency points, enabling the simultaneous and independent probing of individual registers. Hence, IMSCA can compromise security protection schemes based on the *t*-probing security model, such as masking, by recovering all shares simultaneously [13]. Such high-resolution simultaneous probing can only be obtained using more advanced attacks, e.g., laser logic state imaging (LLSI) [58] or thermal laser stimulation (TLS) [59].

5 Conclusions

In CCS'23, it was demonstrated that an adversary could use the impedance or scattering parameters of a CMOS cryptographic device's PDN, implemented on an FPGA, as a side-channel to extract its sensitive information [13]. Given the potential of this side-channel to compromise current countermeasures, accurate measurement of data-dependent impedance, or equivalently *S*-parameter, is critical for researchers and practitioners evaluating their countermeasures against this side channel. Therefore, in this study, we not only evaluated the feasibility of exploiting leakages via IMSC but also conducted a comparative analysis between IMSC and SPSCA. We presented a comprehensive analysis of the effects of temperature and supply voltage as measurement factors that can be controlled to facilitate exploiting leakages. For this purpose, we made use of a 28 nm ASIC prototype chip, which allowed us to experimentally analyze the feasibility of employing the impedance as a side channel in a realistic scenario, i.e., on highly integrated ASIC design. We demonstrated that controlling the temperature can reduce the number of required measurements to detect information leakage through IMSC.

We have observed that electromagnetic radiation (such as the one resulting from the rotating fan of a climate chamber) can significantly affect the measurement results of the scattering parameters. To obtain more precise measurements, it may be beneficial to place this sensitive measurement setup in a Faraday cage. Additionally, according to [34], the accuracy of measurements can be affected by the location of the probe. In our current setup, the PCB, packaging and many other components involved in the path, where the VNA is connected, are not designed for high frequency ranges. Hence, our *S*-parameter measurements at high frequencies have potentially been affected by their inherent low-pass filter. In our future work, we plan to use micro-probes to directly connect the VNA port to the die of the chip and guarantee the bandwidth height in accordance with the frequency of the desired *S*-parameter measurements.

In addition, we observed that temperature variation can change the parasitic elements of the circuit, shifting the measured *S*-parameter across the frequency axis. Therefore, maintaining a constant temperature throughout the *S*-parameter measurement process is essential for the accurate alignment of the recorded measurements. Furthermore, we observed that increasing the supply voltage has a minimal effect on increasing the leakage through IMSC, but increasing the temperature can reduce the number of *S*-parameter measurements required to detect data-dependent leakage.

Finally, since the measurement scenario of impedance SCA and static-power SCA are very similar, we conducted comparative studies between impedance side channel and static power under the same settings. While previous work has demonstrated the feasibility of impedance side-channel attacks on manually placed routed FPGA designs, our research delves deeper into the practical challenges and limitations associated with IMSCA attack under a more realistic scenario, specifically for ASICs. We conclude that in scenarios where static power consumption is high and we can obtain the required measurements with sufficient accuracy, exploiting the static power side channels is much simpler and cost-effective. In short, for unprotected implementations, static-power SCA outperforms impedance SCA by far. Additionally, our findings highlight that although we can increase the amount of information leakage through impedance SCA by controlling the environmental parameters, significant difficulties and expenses are involved when compared to static-power SCA.

While prior works have already demonstrated full key-recovery attacks using IMSCA on FPGA platforms, our ASIC-based experiments reveal that translating such attacks to a highly integrated 28 nm ASIC under realistic measurement constraints is non-trivial. Our MCDPA attempts on the impedance traces did not recover any key material with 40,000 acquisitions per operating point, and extending the campaign to additional frequency bands and more traces would require several months of measurement time per voltage–temperature setting. This underlines that, in the examined scenario, IMSCA is detectable and environmentally controllable but significantly harder to exploit than SPSCA. Estimating Minimum Traces to Disclosure (MTD) presupposes at least one successful key recovery, which was not feasible in our study. Consequently, we report detection metrics, namely t-test, t-statistic growth versus trace count, and time/effort comparisons, and we leave a full IMSCA-based MTD study to future work with broader frequency coverage and higher data acquisition. Our measurements

were conducted on a single 28-nm ASIC. Prior peer-reviewed studies indicate that IMSC leakage depends on register location and routing topology, and impedance profiling is sensitive to layout/structural changes. We therefore expect the frequency regions and effect sizes to vary with design, and we explicitly limit such specific characteristics to our ASIC. Consequently, a multi-design validation study could further investigate potential design-dependent leakage characteristics.

Our ASIC's heterogeneous co-processor set constrains our analysis to a single unprotected design instance. While spatial placement and within-die variability could influence leakage characteristics, exploring these effects would require multiple identical unprotected instances distributed across the die. As a result, potential spatial effects are not characterized in this work and are identified as an important topic for future measurement campaigns on ASICs deliberately designed for such studies.

Further study is required to explore the threat of *dynamic* impedance analysis in scenarios where the circuit is active and switching occurs. In such cases, the overall impedance changes continuously and experiences transient short circuits during CMOS gate switching. An initial analysis reported in [28] shows that dynamic impedance analysis outperforms dynamic power analysis. However, a more thorough study is required to explore the potential of the impedance's dynamic behavior in cases where integration is not possible.

Since IMSC is among the newest side channels, further investigation is required to understand potential exploitation schemes and develop effective countermeasures.

Funding Open Access funding enabled and organized by Projekt DEAL. The work described in this paper has been supported by the European Union through the Twinning Project 101079319 (acronym enCRYPTON).

Declarations

Conflicts of Interests and Competing Interests The authors have no conflicts of interest and no competing interests to declare.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

1. Kocher PC, Jaffe J, Jun B (1999) Differential Power Analysis. In: Advances in Cryptology-CRYPTO '99, 19th Annual International Cryptology Conference, Santa Barbara, California, USA, August 15-19, 1999, Proceedings. Lecture Notes in Computer Science
2. Brier E, Clavier C, Olivier F (2004) Correlation Power Analysis with a Leakage Model. In: CHES 2004. Lecture Notes in Computer Science
3. Moradi A, Standaert F (2016) Moments-Correlating DPA. In: TIS@CCS 2016
4. Mosavirik T, Schaumont P, Tajik S (2023) ImpedanceVerif: On-Chip Impedance Sensing for System-Level Tampering Detection. IACR Trans. Cryptogr. Hardw. Embed, Syst
5. Cnudde TD, Ender M, Moradi A (2018) Hardware Masking. Revisited. IACR Trans. Cryptogr. Hardw. Embed, Syst
6. Giorgetti J, Scotti G, Simonetti A, Trifiletti A (2007) Analysis of data dependence of leakage current in CMOS cryptographic hardware. In: Proceedings of the 17th ACM Great Lakes Symposium on VLSI 2007
7. Genkin D, Shamir A, Tromer E (2017) Acoustic Cryptanalysis. J Cryptol
8. Ferrigno J, Hlavác M (2008) When AES blinks: introducing optical side channel. IET Inf, Secur
9. Hutter M, Schmidt J (2013) The Temperature Side Channel and Heating Fault Attacks. In: Smart Card Research and Advanced Applications - 12th International Conference. Lecture Notes in Computer Science
10. Tiri K, Akmal M, Verbauwhede I (2022) A dynamic and differential CMOS logic with signal independent power consumption to withstand differential power analysis on smart cards. In: European Solid-State Circuits Conference 2002
11. Popp T, Kirschbaum M, Zefferer T, Mangard S (2007) Evaluation of the Masked Logic Style MDPL on a Prototype Chip. In: CHES 2007
12. Nikova S, Rechberger C, Rijmen V (2006) Threshold Implementations Against Side-Channel Attacks and Glitches. In: ICICS 2006
13. Monfared SK, Mosavirik T, Tajik S (2023) LeakyOhm: Secret Bits Extraction using Impedance Analysis. In: Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security, CCS 2023, Copenhagen, Denmark, November 26-30, 2023
14. Nguyen LN, Cheng C-L, Prvulovic M, Zajic AG (2019) Creating a Backscattering Side Channel to Enable Detection of Dormant Hardware Trojans. IEEE Trans. Very Large Scale Integr, Syst
15. Adibelli S, Juyal P, Nguyen LN, Prvulovic M, Zajic A (2020) Near-Field Backscattering-Based Sensing for Hardware Trojan Detection. IEEE Trans Antenn Propagation
16. Jorgensen EJ, Kacmarcik A, Prvulovic M, Zajic AG (2022) Hyperspectral Image Recovery via Reliability-Weighted Compressed Sensing for Hardware Trojan Detection. IEEE Access
17. Nguyen LN, Yilmaz BB, Prvulovic M, Zajic AG (2020) A Novel Golden-Chip-Free Clustering Technique Using Backscattering Side Channel for Hardware Trojan Detection. In: 2020 IEEE International Symposium on Hardware Oriented Security and Trust, HOST 2020
18. Mosavirik T, Monfared SK, Saadat-Safa M, Tajik S (2023) Silicon Echoes: Non-Invasive Trojan and Tamper Detection using Frequency-Selective Impedance Analysis. IACR Trans. Cryptogr. Hardw. Embed, Syst
19. Safa MS, Mosavirik T, Tajik S (2023) Counterfeit chip detection using scattering parameter analysis. In: 2023 26th International Symposium on Design and Diagnostics of Electronic Circuits and Systems (DDECS). IEEE

20. Awal MS, Rahman MT (2023) Disassembling Software Instruction Types through Impedance Side-channel Analysis. In: IEEE International Symposium on Hardware Oriented Security and Trust, HOST 2023, San Jose, CA, USA, May 1-4, 2023
21. Kaji S, Fujimoto D, Kinugawa M, Hayashi Y (2023) Echo TEMPEST: EM Information Leakage Induced by IEMI for Electronic Devices. IEEE Trans Electromag Compatib
22. Digilent (2019) Arty S7™ FPGA Board Reference Manual
23. Moradi A (2014) Side-Channel Leakage through Static Power - Should We Care about in Practice? In: CHES 2014. Lecture Notes in Computer Science
24. Bellizia D, Cellucci D, Stefano VD, Scotti G, Trifiletti A (2017) Novel measurements setup for attacks exploiting static power using DC pico-ammeter. In: ECCTD 2017
25. Pozo SMD, Standaert F, Kamel D, Moradi A (2015) Side-channel attacks from static power: when should we care? In: DATE 2015
26. Moos T, Moradi A, Richter B (2020) Static Power Side-Channel Analysis - An Investigation of Measurement Factors. IEEE Trans. Very Large Scale Integr, Syst
27. Moos T (2019) Static Power SCA of Sub-100 nm CMOS ASICs and the Insecurity of Masking Schemes in Low-Noise Environments. IACR Trans. Cryptogr. Hardw. Embed, Syst
28. Awal MS, Gayanath B, Rahman MT (2024) Impedance vs. Power Side-channel Vulnerabilities: A Comparative Study. arXiv:2405.06242
29. Monfared SK, Forte D, Tajik S (2024) RandOhm: Mitigating Impedance Side-channel Attacks using Randomized Circuit Configurations. In: 2024 IEEE/ACM International Conference on Computer Aided Design (ICCAD). IEEE
30. Bogdanov A, Knudsen LR, Leander G, Paar C, Poschmann A, Robshaw MJB, Seurin Y, Vikkelsoe C (2007) PRESENT: An Ultra-Lightweight Block Cipher. In: CHES 2007. Lecture Notes in Computer Science
31. Smith LD, Bogatin E (2017) Principles of Power Integrity for PDN Design–simplified: Robust and Cost Effective Design for High Speed Digital Products,
32. Mosavirik T, Ganji F, Schaumont P, Tajik S (2022) ScatterVerif: Verification of electronic boards using reflection response of power distribution network. ACM J Emerg Technol Comput Syst (JETC)
33. Zhu Shan H, Sullivan D, Guo X, Jin Y, Zhang X (2023) PDN-Pulse: Sensing PCB Anomaly With the Intrinsic Power Delivery Network. IEEE Trans Inf Foren Secur
34. Smith LD, Sun S, Sarmiento M, Zhe L, Chandrasekar K (2011) On-die capacitance measurements in the frequency and time domains. DesignConm, Santa Clara, CA
35. Pan Z, Cai Y, Tan SXD, Luo Z, Hong X (2004) Transient Analysis of On-Chip Power Distribution Networks Using Equivalent Circuit Modeling. In: 5th International Symposium on Quality of Electronic Design (ISQED, 2004), 22-24 March 2004, San Jose, CA, USA
36. Sotman M, Kolodny A, Popovich M, Friedman EG (2006) On-die decoupling capacitance: frequency domain analysis of activity radius. In: International Symposium on Circuits and Systems (ISCAS 2006), 21-24 May 2006, Island of Kos, Greece
37. Sotman M, Popovich M, Kolodny A, Friedman EG (2005) Leveraging symbiotic on-die decoupling capacitance. In: IEEE 14th Topical Meeting on Electrical Performance of Electronic Packaging, 2005
38. Waizman A, Chung C-Y (2001) Resonant free power network design using extended adaptive voltage positioning (EAVP) methodology. IEEE Trans Adv Packaging
39. MacEachem L, Wang X-J, Kwasniewski TA (2017) On-die power grid broadband model determination using a priori narrowband measurements. In: IEEE 60th International Midwest Symposium on Circuits and Systems, MWSCAS 2017, Boston, MA, USA, August 6-9, 2017
40. Pupalaikis PJ (2020) S-Parameters for Signal Integrity,
41. Pozar DM (2012) Microwave Engineering,
42. Poschmann A, Moradi A, Khoo K, Lim C-W, Wang H, Ling S (2011) Side-Channel Resistant Crypto for Less than 2,300 GE. J. Cryptology
43. Moos T, Moradi A (2021) Countermeasures against Static Power Attacks - Comparing Exhaustive Logic Balancing and Other Protection Schemes in 28 nm CMOS. IACR Trans. Cryptogr. Hardw. Embed, Syst
44. Fadaeinia B, Moos T, Moradi A (2024) Static Leakage in Dual-Rail Precharge Logics. IACR Trans. Cryptogr. Hardw. Embed, Syst
45. Tatarog˘lu A, Altındal S, Bülbül MM (2005) Temperature and frequency dependent electrical and dielectric properties of Al/SiO2/p-Si (MOS) structure. Microelect Eng
46. Razavi B (2021) Fundamentals of Microelectronics,
47. Arora N (2007) Mosfet Modeling for VLSI Simulation: Theory and Practice. Int Series Adv Solid State Electron Technol,
48. Tzou J, Yao C, Cheung R, Chan H (1985) The temperature dependence of threshold voltages in submicrometer CMOS. IEEE Electron Device Lett
49. Groeseneken G, Jean-Pierre Herman, Alderman J, Holt S (1990) Temperature dependence of threshold voltage in thin-film SOI MOSFETs. IEEE Electron Device Lett
50. Tada M, Okamoto K, Tanaka T, Miyamura M, Ishikuro H, Uchida K, Sakamoto T (2024) A 65nm Cryogenic CMOS Design and Performance at 4.2K for Quantum State Controller Application . IEEE J Electron Devices Soc
51. Nelms RM, Johnson RW (1997) 200/spl deg/C operation of a 500-W DC-DC converter utilizing power MOSFETs. IEEE Trans Indust Appl
52. Kar S (2013) MOSFET: Basics, Characteristics, and Characterization,
53. Novkovski N, Atanassova E (2017) Frequency Dependence of Characteristics of MOS Capacitors Containing Nanosized High-κ Ta2O5 Dielectrics. Adv Mater Sci Eng
54. Omar R, Mohamed BA, Adel M (2015) Effects of series and parallel resistances on the C-V characteristics of silicon-based metal oxide semiconductor (MOS) devices. The European Phys J Plus
55. Takahashi T, Matsuki T, Shinada T, Inoue Y, Uchida K (2016) Direct Evaluation of Self-Heating Effects in Bulk and Ultra-Thin BOX SOI MOSFETs Using Four-Terminal Gate Resistance Technique. IEEE J Electron Devices Soc
56. Schneider T, Moradi A (2015) Leakage Assessment Methodology-A Clear Roadmap for Side-Channel Evaluations. In: CHES 2015. Lecture Notes Comput Sci
57. Helms D, Eilers R, Metzdorf M, Nebel W (2018) Leakage Models for High-Level Power Estimation. IEEE Trans. Comput. Aided Des. Integr, Circuits Syst
58. Krachenfels T, Ganji F, Moradi A, Tajik S, Seifert J-P (2021) Real-world snapshots vs. theory: Questioning the t-probing security model. In: 2021 IEEE Symposium on Security and Privacy (SP). IEEE
59. Krachenfels T, Kiyan T, Tajik S, Seifert J-P (2021) Automatic Extraction of Secrets from the Transistor Jungle using Laser-Assisted Side-Channel Attacks. In: 30th USENIX Security Symposium (USENIX Security 21)

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Bijan Fadaeinia received a B.Sc. degree in electrical engineering from the University of Tabriz, Iran, in 2001 and an M.Sc. degree in electrical engineering from the Iran University of Science and Technology, Iran, in 2004. He is currently pursuing a Ph.D. degree with the Chair for Security Engineering, Faculty of Computer Science, Ruhr-Universität Bochum, Germany. From 2004 to 2019, he worked as a Lecturer with the Department of Electrical Engineering at the Islamic Azad University of Hamadan, Iran. His research interests include physical security of embedded devices and hardware-based countermeasures.

Shahin Tajik is an Assistant Professor at the electrical and computer engineering department of Worcester Polytechnic Institute (WPI), Worcester, MA, USA. He received his Ph.D. in Electrical Engineering in 2017 from the Technical University of Berlin in Germany. His field of research includes designing hardware countermeasures and tamper detection mechanisms. He is also the recipient of the NSF CAREER Award.

Amir Moradi received the M.Sc. and Ph.D. degrees in Computer Engineering from the Sharif University of Technology, Tehran, Iran, in 2004 and 2008, respectively, and the Habilitation degree from Ruhr-Universität Bochum, Germany, in 2016. In 2016, he joined the Faculty of Electrical Engineering and Information Technology, Ruhr-Universität Bochum, where he became a Professor, in 2021. From 2021 to 2022, he was with the Institute of Computer Science, University of Cologne, Germany, and from 2022 to 2023, he was with the Faculty of Computer Science, Ruhr-Universität Bochum. Since 2023, he has been working as a Full Professor at the Department of Computer Science, Technische Universität Darmstadt, Germany. His current research interests include physical security of embedded systems, passive and active physical attacks, and the corresponding countermeasures.